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STANS MARGIN ENGINE - MCX Commodity Derivatives Portfolio
OCC STANS : Monte Carlo (10,000 paths) + Historical Simulation (500 paths) | 99% CVaR
Full option revaluation | Correlated joint simulation | Stochastic basis
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Generating correlated return paths via Cholesky decomposition...
MC paths   : 10,000 (GBM + full correlation matrix)
HS paths   : 500 (stress-scaled bootstrapped returns)
Total      : 10,500 paths per position
CVaR level : 99% (tail of worst 1%)

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CORRELATION MATRIX (used in Cholesky joint simulation)
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	Turm	Cori	Jeer	GGum	GSed	Cast	CSOC	Kaps
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Turm	1	0.72	0.68	0.15	0.14	0.1	0.08	0.08
Cori	0.72	1	0.75	0.12	0.11	0.09	0.07	0.07
Jeer	0.68	0.75	1	0.13	0.12	0.08	0.06	0.06
GGum	0.15	0.12	0.13	1	0.82	0.2	0.18	0.18
GSed	0.14	0.11	0.12	0.82	1	0.18	0.16	0.16
Cast	0.1	0.09	0.08	0.2	0.18	1	0.22	0.2
CSOC	0.08	0.07	0.06	0.18	0.16	0.22	1	0.75
Kaps	0.08	0.07	0.06	0.18	0.16	0.2	0.75	1

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STEP 1 : PER-POSITION SIMULATION (10,500 paths | Full Revaluation for Options)
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Row 1 | Turmeric          | Future | Buy × 1 lot
Futures Price : Rs. 16,220.00/Qtl Lot : 5.0 MT Contract Value : Rs. 8,110,000
[STANS] GBM simulation: sigma_daily=1.80% Correlated via Cholesky

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Path Statistics (10,500 total paths):

Metric	Value
-----	-----
Mean Loss	Rs. 314
P1 (best 1%)	Rs. -366,662
P5	Rs. -252,560
P25 (median-ish)	Rs. -100,459
P50 (median)	Rs. 3,125
P75	Rs. 102,205
P95	Rs. 244,006
P99 (VaR 99%)	Rs. 356,465
P99.9 (tail)	Rs. 491,679
Worst path	Rs. 705,440
MC CVaR (99%)	Rs. 386,400
HS CVaR (99%)	Rs. 600,774
Combined CVaR	Rs. 413,114 << MARGIN

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Row 2 | Coriander         | Future | Buy × 2 lot
Futures Price : Rs. 13,350.00/Qtl Lot : 5.0 MT Contract Value : Rs. 13,350,000
[STANS] GBM simulation: sigma_daily=1.80% Correlated via Cholesky

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Path Statistics (10,500 total paths):

Metric	Value
-----	-----
Mean Loss	Rs. 1,105
P1 (best 1%)	Rs. -598,544
P5	Rs. -409,914
P25 (median-ish)	Rs. -164,151
P50 (median)	Rs. 2,452
P75	Rs. 168,342
P95	Rs. 403,557
P99 (VaR 99%)	Rs. 576,817
P99.9 (tail)	Rs. 774,933
Worst path	Rs. 1,081,541
MC CVaR (99%)	Rs. 628,666
HS CVaR (99%)	Rs. 914,746
Combined CVaR	Rs. 674,583 << MARGIN

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Row 3 | Jeera              | Future | Buy × 2 lot
Futures Price : Rs. 22,285.00/Qtl Lot : 3.0 MT Contract Value : Rs. 13,371,000
[STANS] GBM simulation: sigma_daily=2.00% Correlated via Cholesky

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Path Statistics (10,500 total paths):

Metric	Value
Mean Loss	Rs. -1,836
P1 (best 1%)	Rs. -663,449
P5	Rs. -464,079
P25 (median-ish)	Rs. -184,451
P50 (median)	Rs. 3,675
P75	Rs. 188,098
P95	Rs. 443,648
P99 (VaR 99%)	Rs. 622,479
P99.9 (tail)	Rs. 825,915
Worst path	Rs. 973,231
MC CVaR (99%)	Rs. 678,513
HS CVaR (99%)	Rs. 881,522
Combined CVaR	Rs. 705,205 << MARGIN

Row 4 | Guar Gum | Future | Buy × 2 lot
Futures Price : Rs. 10,756.00/Qtl Lot : 5.0 MT Contract Value : Rs. 10,756,000
[STANS] GBM simulation: sigma_daily=1.70% Correlated via Cholesky

Path Statistics (10,500 total paths):

Metric	Value
Mean Loss	Rs. 2,513
P1 (best 1%)	Rs. -435,102
P5	Rs. -314,508
P25 (median-ish)	Rs. -120,651
P50 (median)	Rs. 2,802
P75	Rs. 128,540
P95	Rs. 308,058
P99 (VaR 99%)	Rs. 441,499
P99.9 (tail)	Rs. 647,899
Worst path	Rs. 970,630
MC CVaR (99%)	Rs. 483,793
HS CVaR (99%)	Rs. 845,660
Combined CVaR	Rs. 529,293 << MARGIN

Row 5 | Guar Seed | Future | Buy × 4 lot
Futures Price : Rs. 5,715.00/Qtl Lot : 5.0 MT Contract Value : Rs. 11,430,000
[STANS] GBM simulation: sigma_daily=1.70% Correlated via Cholesky

Path Statistics (10,500 total paths):

Metric	Value
Mean Loss	Rs. 3,161
P1 (best 1%)	Rs. -477,782
P5	Rs. -332,066
P25 (median-ish)	Rs. -130,788
P50 (median)	Rs. 4,683
P75	Rs. 139,198
P95	Rs. 335,552
P99 (VaR 99%)	Rs. 464,665
P99.9 (tail)	Rs. 670,656
Worst path	Rs. 940,674
MC CVaR (99%)	Rs. 508,921
HS CVaR (99%)	Rs. 848,761
Combined CVaR	Rs. 555,666 << MARGIN

Row 6 | Castor Seed | Future | Buy × 4 lot
Futures Price : Rs. 6,392.00/Qtl Lot : 5.0 MT Contract Value : Rs. 12,784,000
[STANS] GBM simulation: sigma_daily=1.60% Correlated via Cholesky

Path Statistics (10,500 total paths):

Metric	Value
Mean Loss	Rs. 4,445
P1 (best 1%)	Rs. -506,527
P5	Rs. -336,023
P25 (median-ish)	Rs. -132,613
P50 (median)	Rs. 5,369
P75	Rs. 144,991
P95	Rs. 339,522
P99 (VaR 99%)	Rs. 478,467
P99.9 (tail)	Rs. 675,515

Worst path	Rs.	890,584	
MC CVaR (99%)	Rs.	525,345	
HS CVaR (99%)	Rs.	813,828	
Combined CVaR	Rs.	567,899	<< MARGIN

Row 7 Cotton Seed OC Future Buy × 2 lot			
Futures Price : Rs. 3,567.00/Qtl Lot : 10.0 MT Contract Value : Rs. 7,134,000			
[STANS] GBM simulation: sigma_daily=1.50% Correlated via Cholesky			
Path Statistics (10,500 total paths):			
Metric		Value	

Mean Loss	Rs.	-956	
P1 (best 1%)	Rs.	-271,294	
P5	Rs.	-182,688	
P25 (median-ish)	Rs.	-72,726	
P50 (median)	Rs.	157	
P75	Rs.	74,239	
P95	Rs.	175,459	
P99 (VaR 99%)	Rs.	248,319	
P99.9 (tail)	Rs.	338,236	
Worst path	Rs.	401,014	
MC CVaR (99%)	Rs.	273,455	
HS CVaR (99%)	Rs.	362,311	
Combined CVaR	Rs.	289,155	<< MARGIN

Row 8 Kapas (Cotton) Future Buy × 7 lot			
Futures Price : Rs. 1,682.00/Qtl Lot : 4.0 MT Contract Value : Rs. 4,709,600			
[STANS] GBM simulation: sigma_daily=1.50% Correlated via Cholesky			
Path Statistics (10,500 total paths):			
Metric		Value	

Mean Loss	Rs.	-814	
P1 (best 1%)	Rs.	-179,545	
P5	Rs.	-122,633	
P25 (median-ish)	Rs.	-49,473	
P50 (median)	Rs.	658	
P75	Rs.	49,555	
P95	Rs.	116,030	
P99 (VaR 99%)	Rs.	163,033	
P99.9 (tail)	Rs.	211,849	
Worst path	Rs.	261,298	
MC CVaR (99%)	Rs.	178,135	
HS CVaR (99%)	Rs.	243,087	
Combined CVaR	Rs.	187,542	<< MARGIN

Row 9 Turmeric Future Buy × 1 lot			
Futures Price : Rs. 16,450.00/Qtl Lot : 5.0 MT Contract Value : Rs. 8,225,000			
[STANS] GBM simulation: sigma_daily=1.80% Correlated via Cholesky			
Path Statistics (10,500 total paths):			
Metric		Value	

Mean Loss	Rs.	319	
P1 (best 1%)	Rs.	-371,861	
P5	Rs.	-256,142	
P25 (median-ish)	Rs.	-101,883	
P50 (median)	Rs.	3,169	
P75	Rs.	103,654	
P95	Rs.	247,466	
P99 (VaR 99%)	Rs.	361,520	
P99.9 (tail)	Rs.	498,652	
Worst path	Rs.	715,443	
MC CVaR (99%)	Rs.	391,879	
HS CVaR (99%)	Rs.	609,293	
Combined CVaR	Rs.	418,972	<< MARGIN

Row 10 Turmeric Future Sell × 1 lot			
Futures Price : Rs. 16,220.00/Qtl Lot : 5.0 MT Contract Value : Rs. 8,110,000			
[STANS] GBM simulation: sigma_daily=1.80% Correlated via Cholesky			
Path Statistics (10,500 total paths):			
Metric		Value	

Mean Loss	Rs.	-314
P1 (best 1%)	Rs.	-356,465
P5	Rs.	-244,006
P25 (median-ish)	Rs.	-102,205
P50 (median)	Rs.	-3,125
P75	Rs.	100,459
P95	Rs.	252,560
P99 (VaR 99%)	Rs.	366,662
P99.9 (tail)	Rs.	516,171
Worst path	Rs.	700,131
MC CVaR (99%)	Rs.	403,350
HS CVaR (99%)	Rs.	584,626
Combined CVaR	Rs.	427,363 << MARGIN

Row 11 | Guar Seed | Future | Buy × 2 lot
Futures Price : Rs. 5,800.00/Qtl Lot : 5.0 MT Contract Value : Rs. 5,800,000
[STANS] GBM simulation: sigma_daily=1.70% Correlated via Cholesky

Path Statistics (10,500 total paths):

Metric	Value
-----	-----
Mean Loss	Rs. 1,604
P1 (best 1%)	Rs. -242,444
P5	Rs. -168,502
P25 (median-ish)	Rs. -66,366
P50 (median)	Rs. 2,376
P75	Rs. 70,634
P95	Rs. 170,271
P99 (VaR 99%)	Rs. 235,788
P99.9 (tail)	Rs. 340,316
Worst path	Rs. 477,332
MC CVaR (99%)	Rs. 258,245
HS CVaR (99%)	Rs. 430,693
Combined CVaR	Rs. 281,965 << MARGIN

Row 12 | Guar Seed | Future | Sell × 2 lot
Futures Price : Rs. 5,715.00/Qtl Lot : 5.0 MT Contract Value : Rs. 5,715,000
[STANS] GBM simulation: sigma_daily=1.70% Correlated via Cholesky

Path Statistics (10,500 total paths):

Metric	Value
-----	-----
Mean Loss	Rs. -1,580
P1 (best 1%)	Rs. -232,333
P5	Rs. -167,776
P25 (median-ish)	Rs. -69,599
P50 (median)	Rs. -2,341
P75	Rs. 65,394
P95	Rs. 166,033
P99 (VaR 99%)	Rs. 238,891
P99.9 (tail)	Rs. 326,351
Worst path	Rs. 394,274
MC CVaR (99%)	Rs. 267,336
HS CVaR (99%)	Rs. 351,277
Combined CVaR	Rs. 276,747 << MARGIN

Row 13 | Jeera | Option | Buy × 1 lot
Underlying : Rs. 22,285.00 Strike : Rs.22,000.00 Call IV_base : 28.0% ? : +0.52
[STANS] Full option revaluation on every path (not B-S approx)
[STANS] Stochastic vol (vol-of-vol=20%) | B-S base px : Rs.865.34

Path Statistics (10,500 total paths):

Metric	Value
-----	-----
Mean Loss	Rs. -2,803
P1 (best 1%)	Rs. -237,830
P5	Rs. -162,289
P25 (median-ish)	Rs. -60,265
P50 (median)	Rs. 3,618
P75	Rs. 61,954
P95	Rs. 134,607
P99 (VaR 99%)	Rs. 175,823
P99.9 (tail)	Rs. 210,279
Worst path	Rs. 232,366
MC CVaR (99%)	Rs. 188,430
HS CVaR (99%)	Rs. 213,344

Combined CVaR Rs. 192,451 << MARGIN

Row 14 | Jeera | Option | Sell × 2 lot
Underlying : Rs. 22,285.00 Strike : Rs.23,000.00 Call IV_base : 28.0% ? : -0.38
[STANS] Full option revaluation on every path (not B-S approx)
[STANS] Stochastic vol (vol-of-vol=20%) | B-S base px : Rs.427.41

Path Statistics (10,500 total paths):

Metric	Value
-----	-----
Mean Loss	Rs. 5,839
P1 (best 1%)	Rs. -213,134
P5	Rs. -172,890
P25 (median-ish)	Rs. -89,326
P50 (median)	Rs. -10,222
P75	Rs. 82,539
P95	Rs. 241,582
P99 (VaR 99%)	Rs. 362,406
P99.9 (tail)	Rs. 562,965
Worst path	Rs. 738,336
MC CVaR (99%)	Rs. 422,213
HS CVaR (99%)	Rs. 683,189
Combined CVaR	Rs. 445,760 << MARGIN

Row 15 | Guar Gum | Option | Buy × 1 lot
Underlying : Rs. 10,756.00 Strike : Rs.10,500.00 Put IV_base : 26.0% ? : -0.48
[STANS] Full option revaluation on every path (not B-S approx)
[STANS] Stochastic vol (vol-of-vol=20%) | B-S base px : Rs.206.40

Path Statistics (10,500 total paths):

Metric	Value
-----	-----
Mean Loss	Rs. -1,472
P1 (best 1%)	Rs. -127,631
P5	Rs. -82,598
P25 (median-ish)	Rs. -29,944
P50 (median)	Rs. 3,303
P75	Rs. 31,518
P95	Rs. 64,287
P99 (VaR 99%)	Rs. 81,670
P99.9 (tail)	Rs. 92,712
Worst path	Rs. 98,519
MC CVaR (99%)	Rs. 86,569
HS CVaR (99%)	Rs. 87,931
Combined CVaR	Rs. 86,650 << MARGIN

Row 16 | Guar Gum | Option | Sell × 1 lot
Underlying : Rs. 10,756.00 Strike : Rs.10,500.00 Put IV_base : 26.0% ? : +0.48
[STANS] Full option revaluation on every path (not B-S approx)
[STANS] Stochastic vol (vol-of-vol=20%) | B-S base px : Rs.206.40

Path Statistics (10,500 total paths):

Metric	Value
-----	-----
Mean Loss	Rs. 1,472
P1 (best 1%)	Rs. -81,670
P5	Rs. -64,287
P25 (median-ish)	Rs. -31,518
P50 (median)	Rs. -3,303
P75	Rs. 29,944
P95	Rs. 82,598
P99 (VaR 99%)	Rs. 127,631
P99.9 (tail)	Rs. 182,997
Worst path	Rs. 296,559
MC CVaR (99%)	Rs. 144,478
HS CVaR (99%)	Rs. 241,640
Combined CVaR	Rs. 153,552 << MARGIN

Row 17 | Castor Seed | Option | Buy × 2 lot
Underlying : Rs. 6,392.00 Strike : Rs.6,200.00 Call IV_base : 24.0% ? : +0.55
[STANS] Full option revaluation on every path (not B-S approx)
[STANS] Stochastic vol (vol-of-vol=20%) | B-S base px : Rs.286.57

Path Statistics (10,500 total paths):

Metric	Value
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Mean Loss                Rs.          194
P1 (best 1%)            Rs.        -204,477
P5                      Rs.        -131,576
P25 (median-ish)        Rs.         -48,709
P50 (median)            Rs.           3,433
P75                    Rs.          54,290
P95                    Rs.         121,795
P99 (VaR 99%)          Rs.         162,893
P99.9 (tail)           Rs.         196,445
Worst path              Rs.         241,409
MC CVaR (99%)          Rs.         173,700
HS CVaR (99%)          Rs.         227,465
Combined CVaR          Rs.    179,719 << MARGIN

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Row 18 | Castor Seed      | Option | Sell × 2 lot
Underlying : Rs. 6,392.00 Strike : Rs.6,200.00 Call IV_base : 24.0% ? : -0.55
[STANS] Full option revaluation on every path (not B-S approx)
[STANS] Stochastic vol (vol-of-vol=20%) | B-S base px : Rs.286.57

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Path Statistics (10,500 total paths):
Metric                                Value
-----
Mean Loss                Rs.          -194
P1 (best 1%)            Rs.        -162,893
P5                      Rs.        -121,795
P25 (median-ish)        Rs.         -54,290
P50 (median)            Rs.          -3,433
P75                    Rs.          48,709
P95                    Rs.         131,576
P99 (VaR 99%)          Rs.         204,477
P99.9 (tail)           Rs.         276,344
Worst path              Rs.         355,302
MC CVaR (99%)          Rs.         225,380
HS CVaR (99%)          Rs.         339,446
Combined CVaR          Rs.    238,868 << MARGIN

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Row 19 | Kapas (Cotton)   | Option | Buy × 3 lot
Underlying : Rs. 1,682.00 Strike : Rs.1,650.00 Put IV_base : 22.0% ? : -0.45
[STANS] Full option revaluation on every path (not B-S approx)
[STANS] Stochastic vol (vol-of-vol=20%) | B-S base px : Rs.28.12

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Path Statistics (10,500 total paths):
Metric                                Value
-----
Mean Loss                Rs.          -302
P1 (best 1%)            Rs.         -40,730
P5                      Rs.         -26,823
P25 (median-ish)        Rs.          -9,378
P50 (median)            Rs.           1,094
P75                    Rs.          10,419
P95                    Rs.          21,057
P99 (VaR 99%)          Rs.          26,933
P99.9 (tail)           Rs.          31,170
Worst path              Rs.          33,352
MC CVaR (99%)          Rs.          28,539
HS CVaR (99%)          Rs.          31,472
Combined CVaR          Rs.    28,972 << MARGIN

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Row 20 | Kapas (Cotton)   | Option | Sell × 3 lot
Underlying : Rs. 1,682.00 Strike : Rs.1,650.00 Put IV_base : 22.0% ? : +0.45
[STANS] Full option revaluation on every path (not B-S approx)
[STANS] Stochastic vol (vol-of-vol=20%) | B-S base px : Rs.28.12

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Path Statistics (10,500 total paths):
Metric                                Value
-----
Mean Loss                Rs.           302
P1 (best 1%)            Rs.         -26,933
P5                      Rs.         -21,057
P25 (median-ish)        Rs.         -10,419
P50 (median)            Rs.          -1,094
P75                    Rs.           9,378
P95                    Rs.          26,823
P99 (VaR 99%)          Rs.          40,730
P99.9 (tail)           Rs.          54,846

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Worst path	Rs.	77,384
MC CVaR (99%)	Rs.	45,855
HS CVaR (99%)	Rs.	61,888
Combined CVaR	Rs.	47,177 << MARGIN

Row 21 | Jeera | Option | Sell × 3 lot
Underlying : Rs. 22,285.00 Strike : Rs.27,000.00 Call IV_base : 28.0% ? : -0.04
[STANS] Full option revaluation on every path (not B-S approx)
[STANS] Stochastic vol (vol-of-vol=20%) | B-S base px : Rs.5.82

Path Statistics (10,500 total paths):

Metric	Value
-----	-----
Mean Loss	Rs. 5,058
P1 (best 1%)	Rs. -5,235
P5	Rs. -5,188
P25 (median-ish)	Rs. -4,364
P50 (median)	Rs. -1,532
P75	Rs. 6,633
P95	Rs. 37,850
P99 (VaR 99%)	Rs. 76,258
P99.9 (tail)	Rs. 159,330
Worst path	Rs. 273,865
MC CVaR (99%)	Rs. 109,141
HS CVaR (99%)	Rs. 168,838
Combined CVaR	Rs. 112,254 << MARGIN

Row 22 | Guar Seed | Option | Sell × 2 lot
Underlying : Rs. 5,715.00 Strike : Rs.4,500.00 Put IV_base : 26.0% ? : -0.03
[STANS] Full option revaluation on every path (not B-S approx)
[STANS] Stochastic vol (vol-of-vol=20%) | B-S base px : Rs.0.07

Path Statistics (10,500 total paths):

Metric	Value
-----	-----
Mean Loss	Rs. 278
P1 (best 1%)	Rs. -75
P5	Rs. -75
P25 (median-ish)	Rs. -70
P50 (median)	Rs. -31
P75	Rs. 180
P95	Rs. 1,608
P99 (VaR 99%)	Rs. 4,629
P99.9 (tail)	Rs. 11,973
Worst path	Rs. 24,624
MC CVaR (99%)	Rs. 7,749
HS CVaR (99%)	Rs. 8,201
Combined CVaR	Rs. 7,808 << MARGIN

Row 23 | Coriander | Future | Sell × 1 lot
Futures Price : Rs. 13,350.00/Qtl Lot : 5.0 MT Contract Value : Rs. 6,675,000
[STANS] GBM simulation: sigma_daily=1.80% Correlated via Cholesky

Path Statistics (10,500 total paths):

Metric	Value
-----	-----
Mean Loss	Rs. -553
P1 (best 1%)	Rs. -288,409
P5	Rs. -201,779
P25 (median-ish)	Rs. -84,171
P50 (median)	Rs. -1,226
P75	Rs. 82,075
P95	Rs. 204,957
P99 (VaR 99%)	Rs. 299,272
P99.9 (tail)	Rs. 427,325
Worst path	Rs. 589,035
MC CVaR (99%)	Rs. 337,556
HS CVaR (99%)	Rs. 502,659
Combined CVaR	Rs. 358,757 << MARGIN

Row 24 | Guar Gum | Option | Buy × 1 lot
Underlying : Rs. 10,756.00 Strike : Rs.8,000.00 Call IV_base : 26.0% ? : +0.91
[STANS] Full option revaluation on every path (not B-S approx)
[STANS] Stochastic vol (vol-of-vol=20%) | B-S base px : Rs.2,756.01

Path Statistics (10,500 total paths):			
Metric	Value		
Mean Loss	Rs.	1,204	
P1 (best 1%)	Rs.	-217,548	
P5	Rs.	-157,253	
P25 (median-ish)	Rs.	-60,350	
P50 (median)	Rs.	1,368	
P75	Rs.	64,272	
P95	Rs.	153,994	
P99 (VaR 99%)	Rs.	220,622	
P99.9 (tail)	Rs.	323,952	
Worst path	Rs.	485,314	
MC CVaR (99%)	Rs.	241,721	
HS CVaR (99%)	Rs.	422,645	
Combined CVaR	Rs.	264,464	<< MARGIN

STEP 2 : CALENDAR SPREAD - Stochastic Basis CVaR (STANS)

[STANS] Both legs simulated jointly; basis = stochastic spread process.

Turmeric: Row 9 (near) <-> Row 10 (far)
 Basis Vol (sigma_basis) : 0.80% (annualised)
 SPAN ICSC charge : Rs.16,100
 STANS Stochastic CVaR : Rs.182,528 (-1033.7% lower)

Guar Seed: Row 11 (near) <-> Row 12 (far)
 Basis Vol (sigma_basis) : 0.60% (annualised)
 SPAN ICSC charge : Rs.11,500
 STANS Stochastic CVaR : Rs.93,536 (-713.4% lower)

Commodity	Near Leg	Far Leg	Basis Vol	SPAN ICSC	STANS CVaR	Reduction
Turmeric	Row 9	Row 10	0.80%	Rs. 16,100	Rs. 182,528	-1033.7% lower
Guar Seed	Row 11	Row 12	0.60%	Rs. 11,500	Rs. 93,536	-713.4% lower

STEP 3 : PORTFOLIO-LEVEL JOINT CVaR (Correlation Diversification Benefit)

[STANS] Full portfolio simulated jointly. Cross-CC correlations reduce net CVaR.

Computing portfolio-level simulation (this may take a moment)...

Sum of individual position CVaRs (no correlation) : Rs. 7,443,936
 Portfolio CVaR (with full correlation matrix) : Rs. 2,345,831
 Diversification Benefit : Rs. 5,098,105
 Diversification Ratio : 68.5%

STEP 4 : CROSS-CC CORRELATION OFFSETS (STANS vs SPAN vs SPAN-2)

Pair	Correlation	SPAN Credit	SPAN-2 Credit	STANS Benefit
Guar Gum <-> Guar Seed	rho=0.82	Fixed 50% table	Corr matrix 44.3%	Full joint sim ~best
Cotton Seed <-> Kapas	rho=0.75	Fixed 40% table	Corr matrix 33.9%	Full joint sim ~best
Coriander <-> Jeera	rho=0.75	None (SPAN skip)	Corr matrix 33.9%	Full joint sim ~best
Coriander <-> Turmeric	rho=0.72	None (SPAN skip)	Corr matrix 30.6%	Full joint sim ~best
Spices portfolio (3-way)	avg rho?0.7	None	Partial	Full 3-way netting *

STEP 5 : POSITION-LEVEL MARGIN SUMMARY (99% CVaR)

Row	Commodity	Type	Side	Qty	Contract Value	Raw CVaR (Rs.)	Margin (Rs.)
1	Turmeric	Future	Buy	1	Rs. 8,110,000	Rs. 413,114	Rs. 413,114
2	Coriander	Future	Buy	2	Rs. 13,350,000	Rs. 674,583	Rs. 674,583
3	Jeera	Future	Buy	2	Rs. 13,371,000	Rs. 705,205	Rs. 705,205
4	Guar Gum	Future	Buy	2	Rs. 10,756,000	Rs. 529,293	Rs. 529,293
5	Guar Seed	Future	Buy	4	Rs. 11,430,000	Rs. 555,666	Rs. 555,666
6	Castor Seed	Future	Buy	4	Rs. 12,784,000	Rs. 567,899	Rs. 567,899
7	Cotton Seed OC	Future	Buy	2	Rs. 7,134,000	Rs. 289,155	Rs. 289,155
8	Kapas (Cotton)	Future	Buy	7	Rs. 4,709,600	Rs. 187,542	Rs. 187,542
9	Turmeric	Future	Buy	1	Rs. 8,225,000	Rs. 418,972	Rs. 91,264
10	Turmeric	Future	Sell	1	Rs. 8,110,000	Rs. 427,363	Rs. 91,264
11	Guar Seed	Future	Buy	2	Rs. 5,800,000	Rs. 281,965	Rs. 46,768
12	Guar Seed	Future	Sell	2	Rs. 5,715,000	Rs. 276,747	Rs. 46,768

13	Jeera	Option Buy	1	Rs.	360,000	Rs.	192,451	Rs.	192,451
14	Jeera	Option Sell	2	Rs.	540,000	Rs.	445,760	Rs.	445,760
15	Guar Gum	Option Buy	1	Rs.	300,000	Rs.	86,650	Rs.	86,650
16	Guar Gum	Option Sell	1	Rs.	240,000	Rs.	153,552	Rs.	153,552
17	Castor Seed	Option Buy	2	Rs.	750,000	Rs.	179,719	Rs.	179,719
18	Castor Seed	Option Sell	2	Rs.	600,000	Rs.	238,868	Rs.	238,868
19	Kapas (Cotton)	Option Buy	3	Rs.	96,000	Rs.	28,972	Rs.	28,972
20	Kapas (Cotton)	Option Sell	3	Rs.	78,000	Rs.	47,177	Rs.	47,177
21	Jeera	Option Sell	3	Rs.	76,500	Rs.	112,254	Rs.	112,254
22	Guar Seed	Option Sell	2	Rs.	60,000	Rs.	7,808	Rs.	7,808
23	Coriander	Future Sell	1	Rs.	6,675,000	Rs.	358,757	Rs.	358,757
24	Guar Gum	Option Buy	1	Rs.	1,050,000	Rs.	264,464	Rs.	264,464

STEP 6 : MARGIN BY STANS COMPONENT

STANS Component	Margin (Rs.)	% Gross
Futures CVaR (Full Simulation)	Rs. 4,281,213	67.8%
Calendar Spread CVaR (Stochastic Basis)	Rs. 276,064	4.4%
Options CVaR (Full Revaluation)	Rs. 1,757,676	27.8%
TOTAL (Gross)	Rs. 6,314,953	100.0%

STEP 7 : MARGIN BY COMMODITY GROUP (CC)

CC Group	Margin (Rs.)	% Total
Spices	Rs. 3,084,652	48.8%
Guar	Rs. 1,690,970	26.8%
Castor	Rs. 986,485	15.6%
Cotton	Rs. 552,847	8.8%

STEP 8 : STANS vs SPAN vs SPAN-2 - METHODOLOGY COMPARISON

Feature	SPAN	SPAN-2	STANS
Risk Measure	Worst of 16 scenarios	Worst of 50 scenarios	99% CVaR (tail avg.)
Simulation	Deterministic grid	Deterministic grid	10,000 MC + 500 HS paths
Option Pricing	B-S (flat vol)	Vol-surface (skew+term)	Full revaluation every path
Stoch. Vol	None	None	Heston-like vol-of-vol=20%
Calendar Spread	Fixed ICSC table	Term-struct. calibrated	Stochastic basis CVaR
Cross-CC Credit	2 pairs (fixed %)	5 pairs (corr. matrix)	Full portfolio netting
Spices Hedge	No credit	rho-based partial credit	Full 3-way Spices netting
Deep ITM Option	B-S only	B-S + gamma nodes	Full path revaluation *
Deep OTM Short	SOM floor	Refined SOM	CVaR replaces SOM *
Gross Margin	Rs.1,63,68,560	Rs.1,56,59,091	Rs.6,314,953
Diversif. Benefit	None	Partial	Rs.5,098,105 *

STEP 9 : APPLY PORTFOLIO DIVERSIFICATION ? NET STANS MARGIN

Sum of Position CVaRs (gross)	: Rs.	6,314,953
Less : Portfolio Diversification Benefit	: Rs.	5,098,105
NET STANS INITIAL MARGIN	: Rs.	1,216,848

STANS FINAL MARGIN SUMMARY

Total Positions	: 24
Futures Positions	: 13
Options Positions	: 11
Total Simulation Paths	: 10,500 per position
CVaR Confidence Level	: 99%
Gross Sum of CVaRs	: Rs. 6,314,953
Portfolio Diversification	: Rs. 5,098,105
NET STANS INITIAL MARGIN	: Rs. 1,216,848
(Reference from workbook	: Rs. 12,19,600 gross)